DATA 240 – Lab Report: Student Portal Log Analytics

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Generated sample submission with dummy data, tables, and figures for PDF extraction evaluation.

This mock assignment analyzes one day of student-portal access logs. The document includes generated dashboards, summary tables, and fabricated incident notes so it can be used as a realistic PDF-to-Markdown test input.

1. Dataset and Goal

The dataset contains timestamp, endpoint, status code, latency, and client region fields. The goal of the analysis is to identify high-volume endpoints, observe latency patterns, and surface short-lived error spikes.

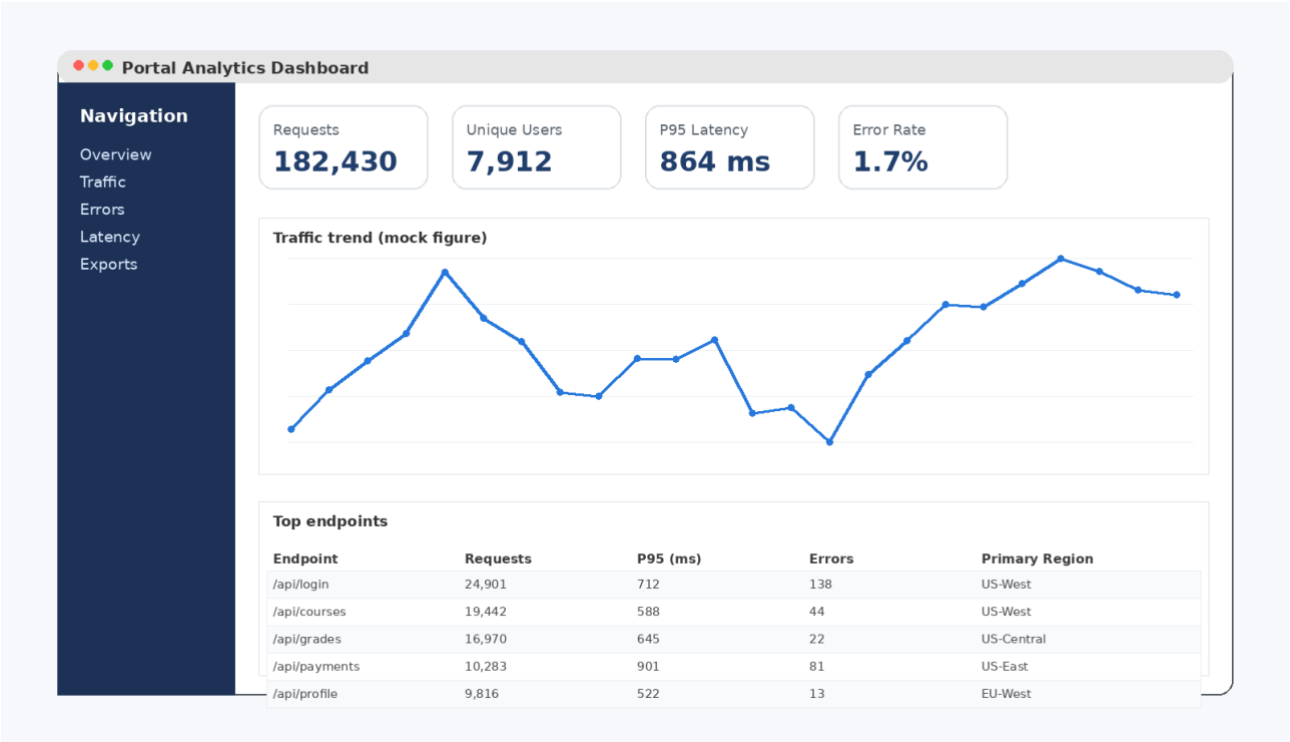

Figure 1. Mock analytics dashboard summarizing portal traffic.

2. Latency and Response-Code Trends

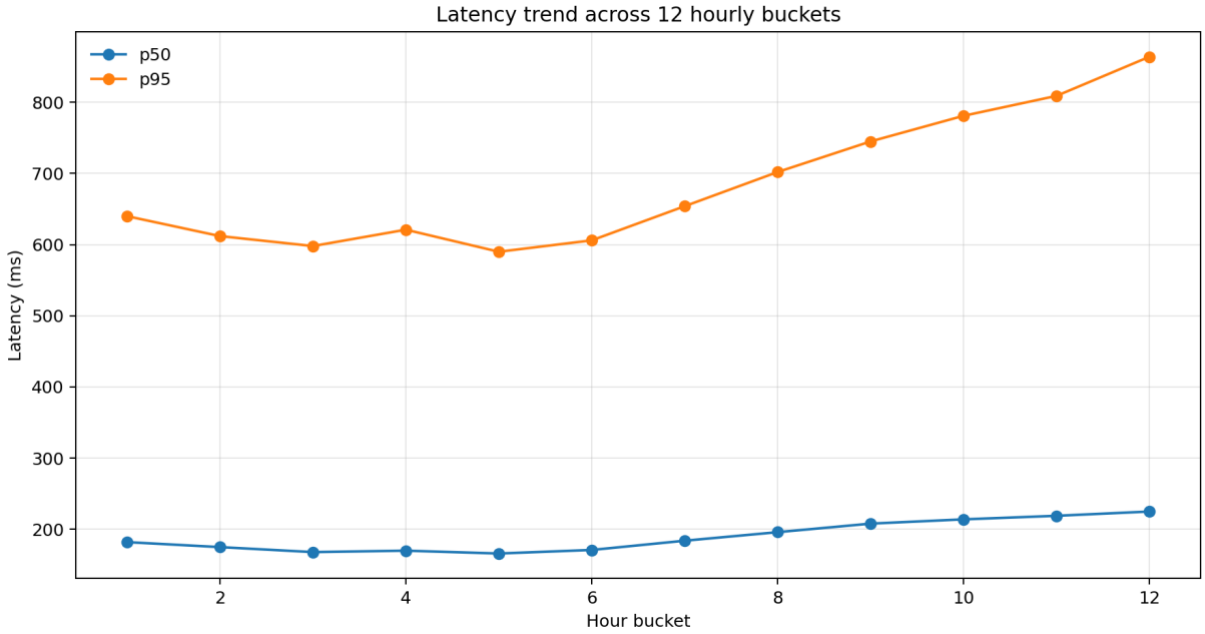

Figure 2. Fabricated latency trend from morning to late evening.

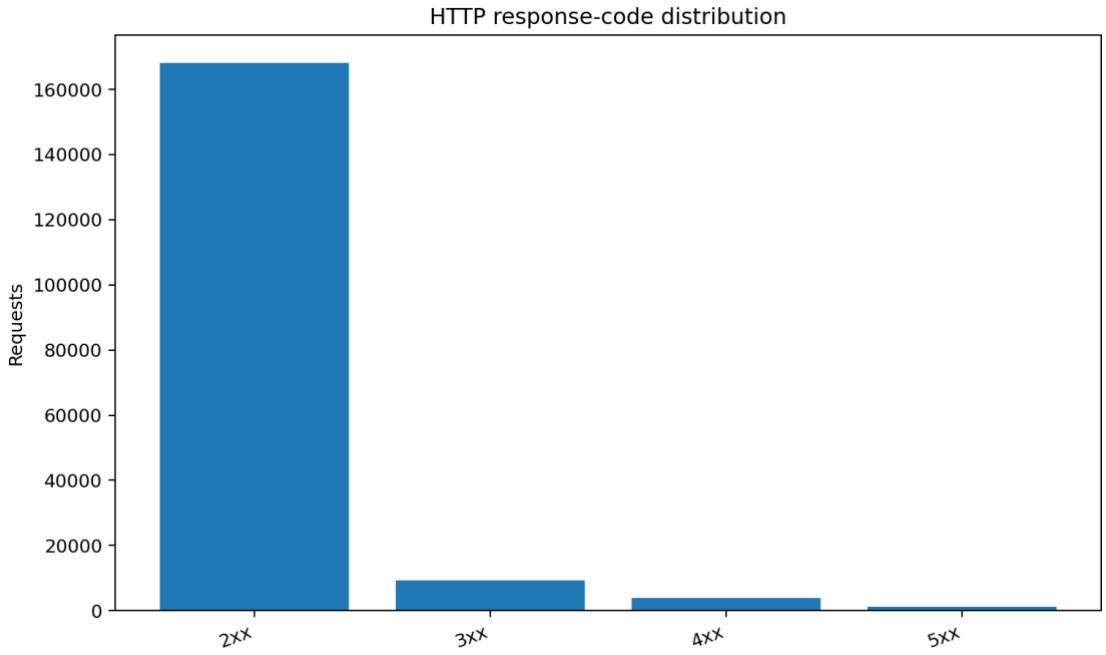

Figure 3. Status-code breakdown for the same observation window.

3. Tabular Findings

Metric	Value	Notes	Severity	Action
Peak request window	7–8 PM	Traffic increased after grade release notifications.	Medium	Scale read replicas before scheduled announcements.
Highest-latency endpoint	/api/payments	Payment callback introduced the largest tail latency.	High	Add timeout budget and async retry handling.
Most common error	401 Unauthorized	Typically triggered by expired mobile sessions.	Low	Prompt token refresh in client app.
Largest error spike	5:20 PM	Short 5xx burst lasted 11 minutes during deployment.	High	Stagger deploys and warm application instances.

4. Incident Notes and Conclusions

1. Traffic patterns were stable for most of the day, but the evening grade-release window amplified both concurrency and latency.
2. The fabricated 5xx spike coincided with an application rollout, which suggests that deployment timing matters as much as raw capacity.
3. Endpoints that depend on external payment services showed the broadest tail, so separate alert thresholds should be used for third-party integrations.
4. Even with dummy data, the report structure approximates the dense mix of screenshots, charts, and tables often seen in student analytics submissions.